Table 1 Least-squares fit for 160.01-min oscillation

		1983	1984	Combined
$f = 104.16\ \mu$Hz				
amplitude	*V*	0.4 ± 0.5	1.7 ± 0.8	0.3 ± 0.4
(marc s)	*D*	0.3 ± 0.5	0.6 ± 0.8	0.3 ± 0.4
Frequency (μHz)	*V*	103.92 ± 0.02	104.23 ± 0.03	104.235 ± 0.003
	D	104.18 ± 0.01	104.00 ± 0.05	104.171 ± 0.003
Amplitude	*V*	1.3 ± 1.3	2.1 ± 2.0	1.7 ± 0.8
(marc s)	*D*	1.8 ± 1.3	1.3 ± 1.9	1.4 ± 0.8

limb exposures and two colours and binned in half-hour intervals. Under these conditions the 'filling factor' on the evenly spaced half-hour domain increases to 23% in 1983 and somewhat less in 1984. The technique we applied to work around the gaps is described by Kuhn[16,17]. By choosing a particular set of evenly spaced frequencies the window function effects can be minimized (the 'swap' technique). Unfortunately, numerical experiments on synthetic data with this domain could not be completely 'dealiased' and we've concluded that the domain is too sparse for this method (and probably any other). We finally settled on the statistical approach, described below, to search for *g*-modes.

We take as a null hypothesis that the 1983 and 1984 power spectra (Fig. 1) are the result of random uncorrelated normally distributed noise. If this is correct then the distribution of power from both years should be consistent with the power distribution generated from noise analysed in the same way as the data. Thus we generate fake data sets on the same domain as the actual data. The values are distributed normally from a distribution whose variance is scaled so that the mean fake power and actual power agree. The resulting cumulative power probability distributions are plotted in Fig. 3. The agreement is quite good if the anamolous peak at 1/6 day period in the 1984 data is ignored. The Kolmogorov–Smirnov test yields a useful quantitative description of the probability that both samples came from the same parent distribution. The K–S test is essentially a measure of the largest deviation between the two distributions[18]. Applying the test to the 1983 and 1984 data separately yields statistic values of 0.34 and 0.86 respectively, indicating that we should accept the null hypothesis. In short, we find no evidence of solar *g*-modes in oblateness power spectra of mean noise 0.4 milliarcsec^2.

Further, if the previously detected 160.01-min periodic signal is interpreted as evidence of an $l = 2$ *g*-mode then its amplitude has decreased from near 1 to 0.1 m s^{-1} since its detection in 1974. Unless a hitherto unknown damping mechanism is postulated this is inconsistent with the expected mode damping time of ~40,000 yr[12]. We believe it is more likely that the 160.01-min signal is not due to a solar *g*-mode.

This research was supported in part by the AFGL under contract number F19628-84-K-0024, and in part by the National Science Foundation.

Received 22 July; accepted 22 October 1985.

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Are the Falkland Islands a rotated microplate?

C. Mitchell*, G. K. Taylor†‡, K. G. Cox* & J. Shaw†

* Department of Earth Sciences, University of Oxford, Parks Road, Oxford OX1 3PR, UK
† Department of Geology, University College, PO Box 78, Cardiff CF1 1XL, UK
‡ Present address: Department of Geology and Physical Sciences, Oxford Polytechnic, Headington, Oxford OX3 0BP, UK

Here we report preliminary palaeomagnetic evidence from dolerite dykes on West Falkland which suggests that the Falkland Islands were rotated through ~120° during the early stages of the break-up and dispersal of the southern part of Gondwanaland. The rotation itself, and evidence of palaeolatitude, confirm the proposal made by Adie[1] in 1952 that the islands were formerly situated adjacent to the Transkei area of South Africa, where they formed the south-east corner of the Karoo basin with its adjoining Palaeozoic fold belts. The first stage of continental drift probably took place during the Jurassic, as Antarctica separated from southern Africa, the islands moving as a microplate to a position approximately 500 km south-east of present-day Cape Town. Subsequently, during the opening of the South Atlantic, the islands and the Falklands Plateau have drifted to their present position and undergone a further rotation of ~60°.

Figure 1 outlines the geology of the Falkland Islands, and the inset shows the position they occupied before the opening of the South Atlantic in the Lower Cretaceous. This position is obtained by closing the South Atlantic by an anticlockwise rotation of the South American plate through 56.40° about a Euler pole at 46.75° N, 32.65° W, keeping the African plate fixed[2]. This is one of the more straightforward plate tectonic reconstructions which is constrained by a wealth of geophysical and bathymetric data, seafloor magnetic anomalies, the orientation of transform faults and the geometrical fit of the opposing continental shelves of South America and Africa[3,4].

Table 1 Palaeomagnetic results

Sample code	Dyke	*n*	*R*	*k*	α_{95}	Dec.	Inc.
F30	A	6	5.62	12.2	19.9	16.0	61.0
F29	A	5	4.86	91.6	8.0	335.4	63.4
F38	A	6	5.97	203.5	4.7	323.1	53.8
F23	B	6	5.92	62.9	8.5	332.1	65.8
F32	A	4	3.70	14.4	25.0	21.8	70.7
F24	B	6	5.93	78.6	7.6	300.5	69.3
F36	C	7	6.79	28.6	11.5	336.3	39.0
F25	B	6	5.94	79.1	7.6	349.2	55.9
F41	D	6	5.89	46.2	10.0	159.5	−24.5*
F31	A	5	4.97	128.9	6.8	7.4	8.6*
F46	E	4	3.94	48.2	13.4	174.9	−54.7
F47	E	6	5.13	5.7	30.6	357.8	−81.5*
F45	E	7	6.10	6.8	25.6	246.1	−24.5*
F6	E	4	—	—	—	—	—*
	Mean	9	8.73	29.5	9.6	342.8	61.1

Palaeomagnetic South Pole = 5.3 °S, 287.2° E. dp/dm = 11.3/14.8, where dp and dm are the semi-axes of the oval of 95% confidence about the palaeomagnetic pole. *n*, The number of specimens; *k*, Fisher's precison parameter; α_{95}, the radius of the 95% confidence cone about the mean direction; dec. and inc., the declination and inclination, respectively. Positive (negative) inclinations indicate reverse (normal) polarity in the Southern Hemisphere. Samples F45 and F47 are total NRM only. The first 10 samples in the table are from four different north–south dykes designated A–D. The last four samples are baked sediments from the contact of an east–west dyke, E.

* Sample direction not included in the calculation of the mean.

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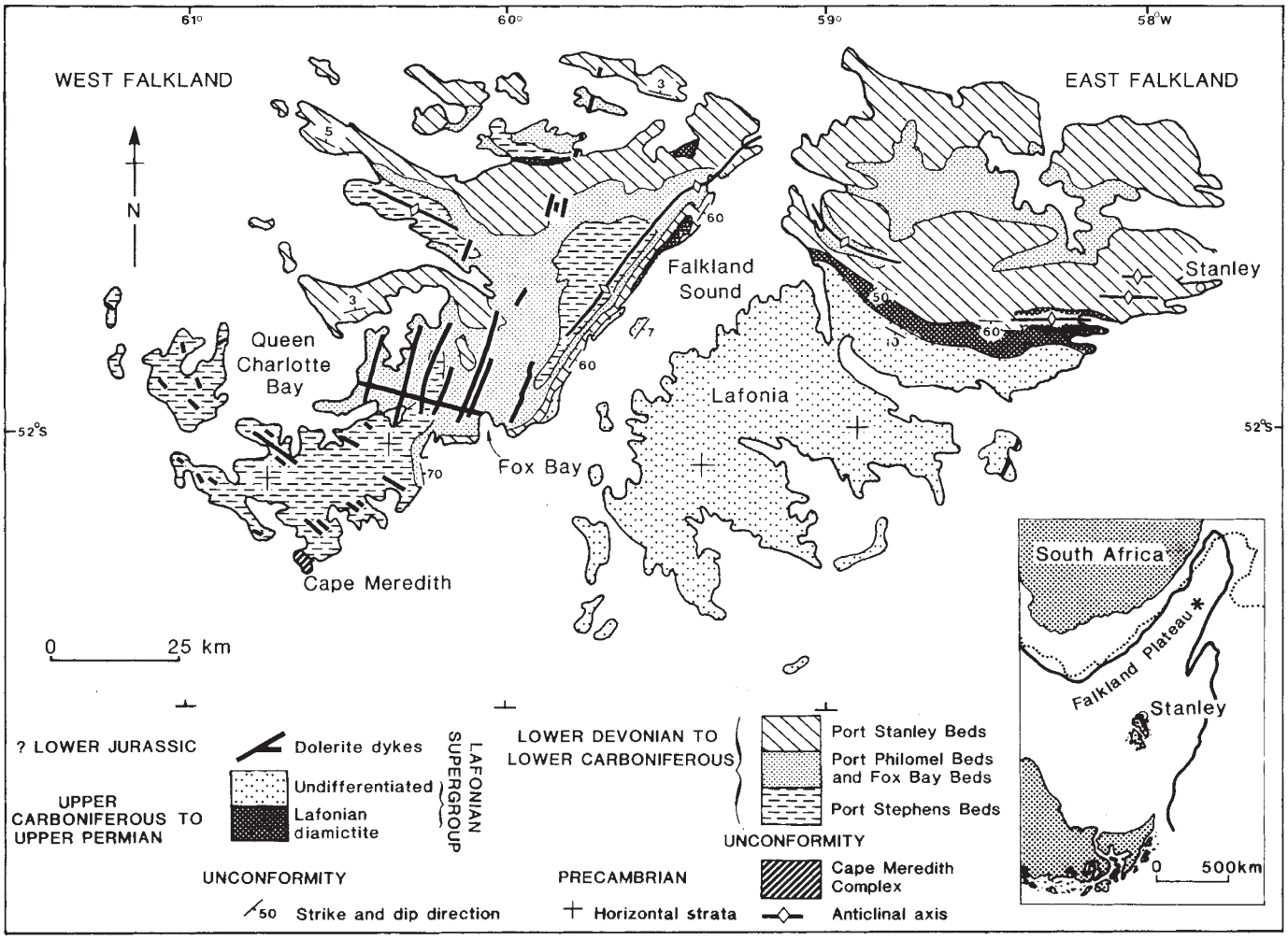


Fig. 1 Geological map of the Falkland Island, after Greenway[17]. For purposes of comparison with South African geology, note that the Lafonian Supergroup correlates broadly with the Karoo, while the Port Stanley Beds, Fox Bay Beds and Port Stephens Beds correlate broadly with the South African units Witteberg, Bokkeveld and Table Mountain sandstone of the Cape Supergroup. Inset, position of the Falkland Islands and Falkland Plateau before the opening of the South Atlantic (Lower Cretaceous) (based mainly on Martin *et al.*[2]). The solid line marks the edge of the south American continental shelf; the dotted line marks that of South Africa. Asterisk shows the position of DSDP site 330 (ref. 14).

Nevertheless, before the development of plate tectonic theory, geologists sought to position the islands in reconstructions of Gondwanaland so that the main geological features matched those of neighbouring continents. For example, du Toit[5] placed the islands to the west of Cape Town, in order to achieve continuity of the east-west fold belts of the Cape, Port Stanley and northern West Falkland in the Falklands and of the Sierra de la Ventana in Argentina. Adie's suggestion[1] that the islands had previously been situated at the east end of the Cape fold belt (the Zwartberg fold belt), offshore from present-day Transkei (Fig. 2), provided near-perfect correlation of Falklands and South African geological features by rotating the islands through 180° from north to south; he cited transport directions in the Dwyka and Lafonian diamictites (with movement from south to north in the Falklands) in support of the postulated rotation. Adie's reconstruction juxtaposes the 1,000-Myr-Natal basement[6] and the Cape Meredith Complex in the Falkland Islands, which is of a similar age[7]; it also removes a problem inherent in the du Toit reconstruction by achieving continuity of the Karoo strata. The Adie reconstruction is much favoured by geological correlations but its validity depends on whether a substantial rotation of the Falkland Islands can be demonstrated.

During a field study in January 1985, C.M. and K.G.C. collected a number of magnetically orientated samples, which comprised 10 hypersthene-bearing dolerites from four different north-south-trending dykes and four baked shales adjacent to a large olivine-bearing east-west dyke, of which, however, no *in situ* exposures were found. All specimens were collected in the area between Fox Bay and Queen Charlotte Bay in West Falkland (Fig. 1). The dykes have usually been assumed to be of the same age as the Karoo dolerites of South Africa; the one published age-determination[8], from a dyke near Cape Meredith, yields a typical Karoo age (Lower Jurassic) of 192 ± 10 Myr by the K-Ar method. Each of our specimens provided a minimum of five 9-mm cylindrical samples for measurement, using the automated SQUID magnetometer at Cardiff[9].

Most samples were demagnetized using alternating fields (AF), but a few were subjected to thermal demagnetization. The results were analysed by means of orthogonal and stereonet plots combined with principal-component analysis[10]. All 10 dolerite samples yielded stable coherent within-sample directions, but two samples (F41 and F31) clearly deviate from the mean of the others (Table 1). These two anomalous results may represent either poor orientation of the samples or the genuine recording of an intermediate geomagnetic field direction. The characteristic direction is generally stable in the range 15–70 mT, with randomly directed low-coercivity components removed on treatment to 15–20 mT, except samples F30 and F32 which were relatively unstable to both AF and thermal demagnetization, compared with the other samples. The AF demagnetization

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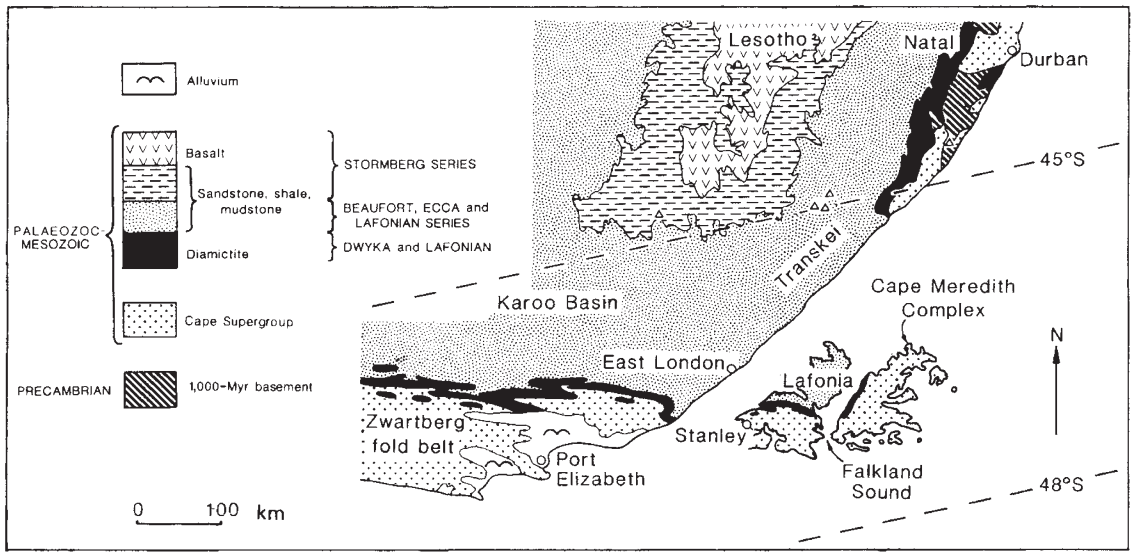


Fig. 2 Proposed reconstruction of the Falkland Islands relative to South Africa in the Lower Jurassic. This reconstruction is essentially similar to that of Aidie (see Fig. 2 of ref. 1). Palaeolatitude lines are based on McElhinny *et al.*[11]. △, Sites from which palaeomagnetic data for Karoo dolerites were obtained.

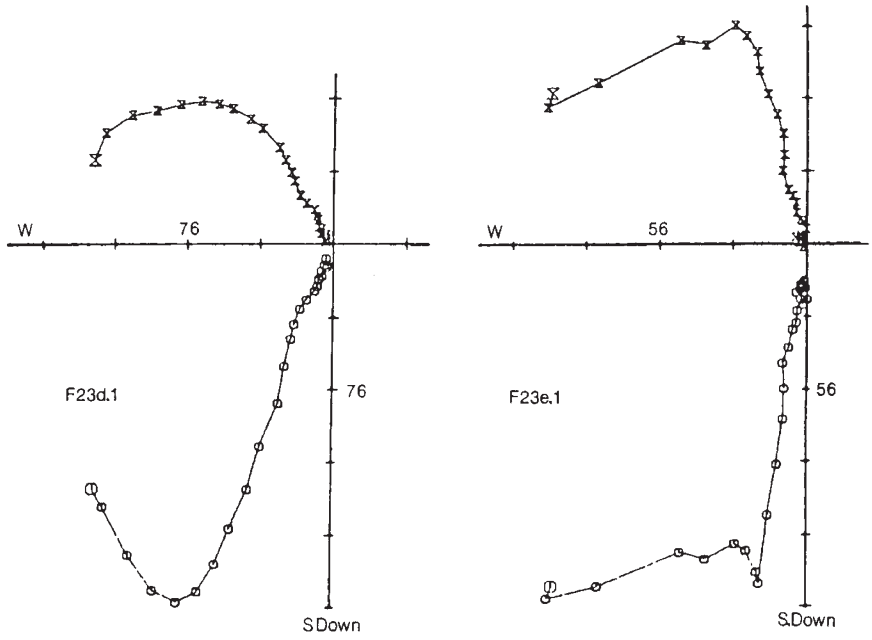


Fig. 3 Orthogonal demagnetization diagrams for two specimens from sample F23. Units for the axes are 10^{-3} Am^{-1}. ×, Horizontal projection; ○, Vertical up-down/east-west projection. Demagnetization steps are in 2.5 mT. See text for discussion.

behaviour of two specimens from sample F23, a hypersthene-bearing north-south dyke, is illustrated in Fig. 3. These two specimens demonstrate the normal range of within-sample variability; in specimen F23d.1 there is overlap of the low-coercivity components, typical of most specimens, while in F23e.1 these components are discrete. In both cases the characteristic direction of the high-coercivity component is readily isolated.

Three of the four baked sediment samples have specimen total NRM (natural remanent magnetism) intensities of $<1.5 \times 10^{-9}$ A m^2 kg^{-1} which are unstable to demagnetization, although samples F47 and F45 do yield poorly defined total NRM directions. Sample F46, however, has specimen total NRM intensities of $2\text{–}5 \times 10^{-8}$ A m^2 kg^{-1} which are stable to AF demagnetization. The AF demagnetization characteristics and thermal stability up to 550–580 °C suggest titanomagnetite as the main magnetic phase present in all samples.

The antiparallel nature of the directions for the dolerites versus the baked sediments (F46, supported by F47 and F45) is indicative of a polarity change between emplacement of the north-south and east-west dykes. The sample mean direction yields an *equatorial* pole position, and this is inconsistent with any data from southern Gondwanaland in Jurassic times. A rotation by a total of $180 \pm 10°$ is needed to produce a palaeo-pole in the expected near-polar position. Further, the palaeolatitude derived from the inclination is $42 \pm 6°$ S, precluding the conventional reconstruction position south of the Cape, which demands a palaeolatitude of ~55 °S. On the other hand, palaeolatitudes derived for southern Africa from Karoo dolerites, admittedly from old and limited data[11] (Fig. 2), suggest a palaeolatitude for the preferred reconstruction position of the Falklands at $47 \pm 5°$ S, which is consistent with our results. However, because of the small number of samples involved, our results are of a provisional nature and will require confirmation by more detailed studies of both Falklands and Transkei dolerites.

Perhaps the most obvious problem with our proposed reconstruction is the mechanism by which the islands have travelled to their present location. All reconstructions of southern Gondwanaland are problematical and authors differ widely in their opinions as to the precise fit. Recent studies[12,13] have indicated that there are several microplates in the area, some

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of which have experienced complex movements, for example the Ellsworth Mountains in Antarctica. We tentatively suggest that the Falkland Islands formed one such microplate which was rotated through 180° and moved south and west as Antarctica moved southward and away from Africa during the Jurassic. The islands were then carried to their present location by the opening of the South Atlantic from the Lower Cretaceous onwards. However, drilling at site 330 (see Fig. 1) on the Maurice Ewing Bank[14] shows that the easternmost segment of the Falkland Plateau is composed of continental crust. Furthermore, Tarney[14] thought it likely that the whole of the Falkland Plateau was composed of similar continental crust because of petrographical and geochemical similarities to the Cape Meredith Complex. However, the crust drilled at site 330 yields an age of 535 Myr[15] hence cannot be correlated with the 1,000-Myr-old gneisses of the Cape Meredith Complex. Also, more recent geophysical surveys[16] of the Falkland Plateau revealed that the Falkland Plateau Basin, which separates the Maurice Ewing Bank from the main plateau, is floored by oceanic or highly thinned continental crust. Ludwig[16] aligned Mesozoic magnetic anomalies off South Africa with their equivalents just off the Maurice Ewing Bank and found that the bank probably occupied a position closer to the plateau than it does at present. He went on to suggest that the Falkland Plateau Basin may represent a zone of seafloor spreading or stretching and subsidence of continental crust during early rifting of the continents. Therefore, it now seems unlikely that the Falkland Plateau is a simple extension of the South American and South African continental crust, and we suggest that it may be a collage of microplates. If, as might be expected, all the microplates recognized in this area experienced complex motions during the break-up and dispersal of southern Gondwanaland, this may explain why reconstructions of southern Gondwanaland are so problematical.

We thank Dr C. W. M. Swithinbank (British Antarctic Survey) for drawing our attention to the existence of the Falklands dolerites. Thanks are also due to Mr and Mrs A. McMullen for their hospitality at Fox Bay West. Field and laboratory work in Oxford were supported by NERC research grant GR3/5350 and by a travel grant from the Royal Society. NERC grant GR3/3901 supported the development of the super conducting magnetometer used for this work at Cardiff.

Received 12 July; accepted 28 October 1985.

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Production of ^{10}Be and ^{26}Al by cosmic rays in terrestrial quartz *in situ* and implications for erosion rates

K. Nishiizumi*, D. Lal†, J. Klein‡, R. Middleton‡ & J. R. Arnold*

*Department of Chemistry, B-017, and †Scripps Institute of Oceanography, A-020, University of California, San Diego, La Jolla, California 92093, USA
‡Tandem Accelerator Laboratory, Department of Physics, University of Pennsylvania, Philadelphia, Pennsylvania 19104, USA

We present results of determinations of ^{10}Be (half life = 1.6 Myr) and ^{26}Al (half-life = 0.705 Myr) produced by cosmic rays *in situ* in several terrestrial rock samples exposed at altitudes of 1–4 km. These experiments were designed to determine the feasibility of using quartz extracted from natural surfaces for studying continental weathering/erosion processes[1]. Quartz is attractive as a target material for several reasons. Its low aluminium concentration allows measurement of ^{26}Al after exposures as short as 10^3 yr; its integrity minimizes the possibility of contamination by rain with ^{10}Be produced in the atmosphere; and its ubiquity ensures that it is found in a wide variety of geological settings. However, the cosmic-ray production rates of ^{26}Al and ^{10}Be even at mountain altitudes are low[2,3], only 10–100 atoms g^{-1} yr^{-1}, with the consequence that earlier attempts[4,5] to determine *in situ* produced ^{36}Cl and ^{26}Al in rocks were not continued. Recent development of accelerator mass spectrometry (AMS) however, has made the measurement of 10–20g samples feasible. The results presented here demonstrate the feasibility of quantitatively measuring ^{10}Be and ^{26}Al produced *in situ* by cosmic rays in quartz and the possible applications of ^{10}Be and ^{26}Al as a pair for studying continental weathering/erosion processes.

Table 1 Sample description

Sample	Site	Elevation (m)	Weight of quartz analysed (g)	Al content of quartz (p.p.m.)	Be carrier added (mg)
ALH	Allan Hills, Antarctica	2,050	13.09	467	1.49
Q-1	Anza Borrego, California	1,140	20.22	4.9×10^4	1.37
Q-L	Anza Borrego California	1,200	22.18	100	1.50
KKS-1	Leh, Ladakh	4,000	17.29	52*	1.50
KKS-3	Kinnaur, Himachal Pradesh	3,400	16.39	289	1.50

* Intrinsic Al content of the purified quartz grains; an amount equivalent to 0.53 mg Al was added for AMS measurement.

The rock samples analysed are listed in Table 1. Sample ALH, which is quartzite (determined by thin section), was collected at the peak of Allan Hills, Antarctica (76°42′ S, 150°31′ E), elevation 2,050 m, by one of us (K.N.) during the 1983–84 field season of Antarctic search for meteorites. Allan Hills are located about 230 km north-west of McMurdo station. The large blue ice area (Allan Hills ice field) is stagnated by the Allan Hills. More than 2,000 meteorites have been found on the ice field and contiguous areas during the past 10 yr. At present, the Allan Hills are not covered with ice because of strong winds which prevent net snow accumulation in winter; in summer there is no precipitation. It would be useful to determine when Allan Hills lost their ice sheets. The sample was collected from the surface of a flat-top area, ~200 m north of the glaciological survey station[6]. The Q-1 and Q-L samples were collected from Anza Borrego, California (33°12′ N, 116°27′ W), ~100 km east of San Diego by A. E. Engel and us. The samples were collected from the flat tops of large rocks (~5 m in diameter) located at

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